

Program: Diploma in Microelectronics	
Course Code: 5492	Course Title: Microcontroller and Applications
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To equip students on architecture, interfacing and Assembly Language Programming (ALP) of 8051 microcontrollers.
- To enable students to develop simple interfacing applications for daily use
- To prepare students for troubleshooting of microcontroller-based equipment.

Course Prerequisites:

Topic/Description	Course code	Course Title	Semester
Basic Engineering mathematics principles and theorems		Mathematics I & II	1 & 2
Different number systems, codes, combinational and sequential circuits, memory		Digital Electronics	3

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the features and block diagram of 8051 Microcontroller.	12	Understanding
CO2	Develop simple Assembly Language Programs (ALP) for 8051.	17	Applying
CO3	Explain interrupts, timer and serial communication in 8051	14	Understanding
CO4	Illustrate the interfacing of various I/O devices with 8051	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the features and block diagram of 8051 Microcontroller.		
M1.01	Compare Microprocessor and Microcontroller	2	Understanding
M1.02	List the features of 8051 microcontroller.	1	Remembering
M1.03	Explain the block diagram of 8051 microcontroller.	2	Understanding
M1.04	Explain the memory of 8051.	7	Understanding
Contents: Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.			
CO2	Develop simple Assembly Language Programs (ALP) for 8051.		
M2.01	Explain the structure of Assembly Language Programming.	3	Understanding
M2.02	Explain various addressing modes and types of instructions.	4	Understanding
M2.03	Describe Jump, Loop, Call and single bit level instructions.	4	Understanding
M2.04	Develop Assembly language programs for 8051.	6	Applying
	Series Test – I	1	

Contents:

Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.

CO3	Explain interrupts, timer and serial communication in 8051.		
M3.01	Explain concept, types, priority, Registers for Interrupts and handling of interrupts.	4	Understanding
M3.02	Explain TMOD register and timer operating modes.	4	Understanding
M3.03	Illustrate TCON register and time delay calculation for Timer.	3	Understanding
M3.04	Illustrate special function registers for serial communication.	3	Understanding

Contents:**Interrupts, Timers, Serial communication.**

8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register.

Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation.

Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register.

CO4	Illustrate the interfacing of various I/O devices with 8051		
M4.01	Explain the interfacing of stepper motor with 8051.	4	Understanding
M4.02	Explain the interfacing of DC motor for speed control.	3	Understanding
M4.03	Illustrate the interfacing of 4x4 keyboard and 16x2 LCD system with 8051.	4	Understanding
M4.04	Illustrate ADC, DAC and LM35 temperature sensor interfacing with 8051.	4	Understanding
	Series Test – II	1	

Contents:

Stepper motor – operation, explain the interfacing with 8051

DC motor – operation, explain the interfacing with 8051

4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051.

16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051.

Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor.

Text / Reference:

T/R	Book Title/Author
T1	Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller and Embedded Systems Using Assembly and C - Pearson Education- Second Edition.
T2	Subratha Ghoshal – 8051 microcontroller internals, instructions, programming and interfacing- Pearson
T3	Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition.

Online Resources:

Sl.No	Website Link
1	https://www.electronicshub.org
2	https://www.elprocus.com
3	https://www.tutorialspoint.com